Luis A. Rodriguez Condit

Technical Architect – Embedded & Web Systems

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Summary

Software architect and developer with over 20 years of software development experience, specializing in bringing complex, low-level systems to modern web applications that have been used by millions. My languages of choice are JavaScript, TypeScript, Java, and C/C++. I thrive owning solutions from end to end, a proven technical leader, skilled in translating business needs into user and technical requirements, driving code quality by advocating for clean and maintainable practices, and mentoring teams to deliver robust, performant, and user-centric solutions.

Core Technical Skills

Languages & frameworks	TypeScript, JavaScript, Angular/React, C/C++, Java, Z80/eZ80 assembly
Systems & low-level	Hardware emulation (Z80, eZ80, T4, Lapis MCU), embedded systems, control systems, concurrent programming, communication protocols
Frontend & UI/UX	WebAssembly, HTML5, CSS, cross-browser/platform UI, WebUSB/WebSerial integration, accessibility
DevOps & testing	Git, CI/CD (Jenkins/Gulp/CMake/MSBuild), build automation, Docker, Mocha/Chai/Jasmine, Jest/React Testing Library, Selenium, Jira/Confluence/Bitbucket
Architecture & backend	RESTful services, SPA architecture, Node.js, object-oriented design, OpenAPI/Swagger
Leadership	Technical leadership, mentorship, Agile/Scrum, test-driven development

Professional Experience

Texas Instruments -- Software Architect -- *(Nov 2010 – Sep 2025, Dallas, TX)*

Built desktop and web applications that bridge low-level hardware emulation and proprietary communication protocols, bringing embedded systems (calculators) to millions of users. Led technical planning and project execution for multiple development teams. Recognized with appointment to TI's Technical Ladder for technical excellence.

Web Apps

- Architected and led development of multiple SPAs using TypeScript and Angular, serving over 3.5 million users globally.
- Defined front-end architecture across calculator application iterations, ensuring responsive, cross-platform experiences.
- Translated complex product requirements into user-centric interfaces, partnering with product management and UX designers.
- Built RESTful services and APIs enabling license activations and data synchronization between Angular frontend and backend systems.
- Owned the development lifecycle from planning through delivery—translating features into actionable Jira epics, leading Scrum execution, and coordinating across QA and product teams to ship on schedule.

Hardware Emulation

- Built high-performance calculator emulators for the browser, implementing Z80, eZ80, Toshiba T4, and Lapis MCU architectures in pure JavaScript for maximum browser compatibility.

- Led WebAssembly adoption for a major calculator platform, cross-compiling C code to run natively in the browser and significantly improving performance.
- Implemented accessibility support across Z80 assembly and JavaScript layers, enabling screen reader integration for calculator emulators.

Communications

- Integrated WebUSB and WebSerial APIs into web applications, enabling direct browser-to-calculator communication for TI-Nspire Connect and other platforms.
- Enabled test automation for calculator emulators through a WebSocket-driven harness that allowed automated tests (Robot Framework + Python) from other platforms to be reused.

Proof of Concept Development and Prototypes

- Accelerated technology adoption by delivering rapid prototypes and implementations, including Node.js automation frameworks with Mocha and Selenium.
- Reduced build times by 70% through Jenkins parallelization, Gulp, CUnit and CMake optimization.
- Raised code quality standards across multiple teams through code reviews and mentorship of junior and mid-level engineers.

Digital Chocolate -- Game Engineering Manager -- *(Jun 2009 – Oct 2010, Mexicali, Mexico)*

- Managed teams responsible for porting mobile games across various US carrier devices, focusing on technical quality and resource allocation. Mentored the engineering team through hands-on tutorials and code deep-dives, directly contributing to team skill development.
- Created a client module for Qualcomm's Application Value Billing API (C/C++) to enable in-app purchases and promote code reuse across future game titles.

LemonQuest -- iPhone Programmer -- *(Apr 2008 – Jun 2009, Salamanca, Spain)*

- Ported games from DirectX to iPhone's OpenGL ES, optimizing the user experience by translating desktop inputs for touch and accelerometer use.

Gameloft -- Game Programmer -- *(Aug 2005 – Mar 2008, Mexicali, Mexico)*

- Led the team that developed the first successful iPhone game prototype for the studio, demonstrating rapid feature development and adaptability to emerging mobile/browser platforms.
- Led the technical effort to port multiple game titles to resource-limited, 3D-capable mobile devices using Java/J2ME and OpenGL ES, focusing on maintaining graphic fidelity and stability within strict memory constraints.
- Directed and participated in team efforts for porting games to severely space-constrained devices (Virgin Mobile catalog). This required strategic scope reduction, asset compression, and intensive micro-optimizations to ensure acceptable performance and quality within limited runtime memory.

Education & Professional Development

Centro de Investigación Científica y de Educación Superior de Ensenada (CICESE)

- Completed coursework towards: Master's in Electronics and Telecommunications
  - Developed RT-Linux kernel modules for low-latency robot arm control, implementing TCP communication between PD control algorithms and motor drivers. Built Java Swing UI with JNI integration and kernel-space I/O for direct data acquisition card access.

Instituto Tecnológico y de Estudios Superiores de Monterrey (ITESM)

- Bachelor's in Electronics Systems Engineering

Professional Development: Online Courses

- In Progress: Self-study and projects focused on Docker containerization React with Vite front end with HAProxy, Spring Boot/nginx and PostgreSQL (Q4 2025).
- Selected courses: React Fundamentals, JavaScript Design Patterns, Web Performance, Software Testing, AI/Robotics.

DISCLAIMER: Portions of this document were created using an AI to tailor experience points to the target role. All content, including dates, titles, and technical claims, is 100% accurate and based on my verified career history. I have thoroughly proofread and confirmed its factual correctness.